The empirical recurrence for OEIS A252141, recovered from its tail

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Abstract

OEIS A252141 records "Empirical recurrence of order 96" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 283$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 10$, so the recurrence is proved for every $n \geq 106$. Its last coefficient is $c_{96} = -4 \neq 0$, so no further tail satisfies anything shorter; any order-96 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A252141 is "Number of (n+2)X(1+2) 0..3 arrays with every 3X3 subblock row and column sum equal to 0 3 5 6 or 7 and every 3X3 diagonal and antidiagonal sum not equal to 0 3 5 6 or 7.". It has offset 1 and begins

$$1921, 3026, 5090, 8132, 14578, 24855, 44707, 79081, \ \ldots$$

The entry states:

> Empirical recurrence of order 96 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 10:51:01, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \; = \; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 283$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 283$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 96: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 10$ is the first whose minimal order is exactly the stated 96.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 566$ exact terms from $n = 10$ on, it returns order 96 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{96} c_i\, a(n-i),$$

c_1	0	c_{25}	−22244	c_{49}	58372	c_{73}	−28368
c_2	6	c_{26}	−55633	c_{50}	−121237	c_{74}	−17705
c_3	16	c_{27}	−30657	c_{51}	−201036	c_{75}	2932
c_4	−9	c_{28}	27071	c_{52}	−89570	c_{76}	14083
c_5	−88	c_{29}	47707	c_{53}	76386	c_{77}	10735
c_6	−118	c_{30}	10373	c_{54}	116006	c_{78}	1233
c_7	115	c_{31}	−20326	c_{55}	43142	c_{79}	−5266
c_8	567	c_{32}	−814	c_{56}	−9630	c_{80}	−5345
c_9	537	c_{33}	29332	c_{57}	−11611	c_{81}	−2125
c_{10}	−578	c_{34}	−2377	c_{58}	−24872	c_{82}	876
c_{11}	−2062	c_{35}	−80442	c_{59}	−45607	c_{83}	1817
c_{12}	−1697	c_{36}	−92614	c_{60}	−28957	c_{84}	987
c_{13}	1314	c_{37}	20076	c_{61}	6139	c_{85}	−50
c_{14}	4339	c_{38}	164528	c_{62}	32878	c_{86}	−386
c_{15}	3663	c_{39}	165563	c_{63}	52478	c_{87}	−131
c_{16}	−519	c_{40}	4941	c_{64}	50181	c_{88}	154
c_{17}	−3616	c_{41}	−145808	c_{65}	−2560	c_{89}	86
c_{18}	−3758	c_{42}	−146972	c_{66}	−69282	c_{90}	−78
c_{19}	−4558	c_{43}	−53193	c_{67}	−71988	c_{91}	−90
c_{20}	−7886	c_{44}	−5222	c_{68}	−11095	c_{92}	−17
c_{21}	−5032	c_{45}	−3914	c_{69}	44413	c_{93}	34
c_{22}	13405	c_{46}	33022	c_{70}	50905	c_{94}	14
c_{23}	33119	c_{47}	122361	c_{71}	20617	c_{95}	2
c_{24}	23201	c_{48}	161787	c_{72}	−13997	c_{96}	−4

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 106$, and it is the recurrence OEIS A252141 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 10$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{96} - c_1 x^{96-1} - \cdots - c_{96}$. Its constant term is $-c_{96} = 4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 96 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 96, so $\deg q = 96$, and it is monic. It holds from some index t on. From $\max(t, 10)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 96, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 26 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 96, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -4.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-96 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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